

Charalampos Lazaridis Patsalias

AI Researcher · sequence-model architectures, training objectives & interpretability

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Looking for

Funded PhD position or
research role in core AI
Also open to
research-engineer roles
Anywhere in Europe

Research interests

Development of AI
architectures and objectives
AI interpretability
World models
Self-supervised learning
and scaling

Languages

Greek | native
English | C1+

Skills

Research methods

Transformers, Mamba / SSMs,
JEPA, self-supervised learning;
linear & MLP probing; causal
interventions; factorial design,
multi-seed evaluation,
bootstrap CIs

ML stack

Python, PyTorch, TensorFlow /
Keras, scikit-learn, CatBoost,
Optuna, SHAP; imbalanced
classification, feature selection,
ensembles

LLM systems

LangChain / LangGraph, RAG,
vector databases, structured
outputs, LLM-as-a-judge,
async orchestration, Hugging
Face Hub

Cheminformatics

RDKit, Mordred, AutoDock
Vina, ADME & toxicity
modelling

Engineering

SQL, C++ (working knowledge),
FastAPI, Flask, Docker (basic),
OpenCV; reproducible
pipelines (SHA-256-linked
checkpoints, data manifests)

ABOUT

AI/ML researcher with a theoretical-physics background and over five years of industry experience in biotechnology and statistical simulation. Developed predictive models for molecular properties, automated toxicology workflows, and built LLM-based systems for scientific knowledge extraction and compound prioritisation. Pipelines over **100k+ papers** narrowed thousands of candidates to **50 for lab testing**; asynchronous LLM workflows reduced report generation from **15 to 2 minutes**. At GLI, assessed **150+ games in total** against jurisdiction-specific requirements through game simulations and statistical analysis. Recent MSc research extends this experimental background to **core AI**, investigating how architecture and training objectives shape internal representations in sequence models. Through iterative experimentation, developed a **JEPA-like contrastive objective using InfoNCE** that, when paired with Mamba, rivalled autoregressive training in legal-move prediction and surpassed it in linear board-state decoding. Evaluated Transformers and Mamba through behavioural tests, probing and causal interventions. Seeking a funded PhD or core AI research role.

RESEARCH

2024–2026

The Emergence of Internal Representations in Sequence Models

MSc DISSERTATION, UNIVERSITY OF ESSEX

Supervisor: Athanasios Chasapis · Oct 2026 · 102 pp.

- **Question & design.** Does the Othello world-model finding (Li et al.; Nanda et al.) generalise beyond one Transformer at one scale? $2 \times 2 \times 3$ factorial study: Transformer vs Mamba $\times$ autoregressive vs JEPA-like objective $\times 8 \times 8, 12 \times 12, 16 \times 16$ boards. **12 conditions**, shared data, parameters matched within **0.86%**, one evaluation protocol, **5 seeds** at the extreme scales; every result linked to its checkpoint via SHA-256, with versioned data manifests and locked library versions.
- **Three levels of “understanding”.** Behaviour (top-1 legality, P@k, legal probability mass); decodability (linear and MLP probes at every layer, board relative to the player to move); causal use (three intervention protocols against a geometry-matched random control).
- **Built a JEPA objective for discrete sequences.** Traced a five-version failure to the predictor exploiting the shared prefix of nested views, and to a supervision asymmetry (one vs all positions); hard-disjoint, all-position targets lifted top-1 legality from **25.4% to 99%+** (dev runs, 8×8).
- **Finding 1: the objective's effect depends on the architecture.** JEPA improves Mamba's board representation and degrades the Transformer's: interaction **+5.09 pp** at 16×16 (95% CI +4.23 to +5.95, 5 seeds). Mamba-JEPA decodes the board best at every size (**83.77%** at 16×16) despite never training on next-move prediction.
- **Finding 2: the standard benchmark hides this.** The best-to-worst spread grows from **1.32 to 5.39 pp** with board size, while top-1 legality stays above 97.5%.
- **Finding 3: the represented board is causally used, less so at scale.** The adapted intervention raises probability mass on the edited board's legal moves in **all 12 conditions** and beats the random control in all 12; the effect shrinks from **5.97–6.37 pp** at 8×8 to **0.78–1.48 pp** at 16×16 .

📄 Full thesis 📄 2-page summary

INDUSTRY EXPERIENCE

Jan 2022 –
Aug 2025

ML & AI Researcher

SIFTLINK (BIOTECH STARTUP) · Remote 📍

- **Compound prioritisation.** Combined LLM extraction over 100k+ PubMed papers, LLM-as-a-judge validation, Lipinski filtering and AutoDock Vina docking to cut thousands of candidates to a **50-compound shortlist** for lab testing; **compound #3 was selected for development**.
- **Bioavailability prediction.** 1,800+ molecular descriptors per compound, Elastic Net feature reduction and an ensemble with class-probability ranking on imbalanced data: **ROC-AUC 0.85+**; the ranking decided which compounds got bench time.
- **Scientific RAG & knowledge extraction.** RAG pipelines and an LLM extraction agent over papers, patents and biomedical sources (embeddings, vector DB, retrieval against a Chemical/Gene/Protein/Pathway ontology, structured outputs); Lang-Graph agent with tool calling and intent-based routing; every extracted entity validated before reporting.
- **Evaluation.** LLM-as-a-judge scoring of output validity (0–100) against the retrieved context, to compare prompts, retrieval strategies and models systematically.
- **Latency.** Replaced sequential LLM calls with asynchronous execution: report generation from **15 to 2 minutes (–87%)**.
- **Toxicology & ADME.** Automated training and inference for **616 toxicological models** plus ADME property models for systematic compound-safety screening.
- **Computational biology.** Salt-taste mechanisms (literature RAG, knowledge graphs, docking); review of flavour-prediction data and models.

Links

Live demo (HF Spaces) 
 Demo source code 
 Portfolio: clazaridis.com 

Teaching

Teaching Assistant,
 Mathematical Analysis
 Maastricht University,
 2017–2018

Writing

MSc thesis (102 pp.)
 Research summary (2 pp.)
 3 project reports (22–38 pp.)
 all in English, at clazaridis.com

EARLIER EXPERIENCE

- 2020–2021 **Retraining into data science**
 ONLINE COURSES & CERTIFICATES
 Systematic coursework in data science and predictive modelling, closing with the IBM Data Science capstone (Dec 2021).
- Oct 2018 – Jul 2020 **Mathematical Analyst**
 GLI EUROPE · Haarlem, Netherlands 
 Ran game simulations, collected outcomes and performed statistical analyses to assess compliance with jurisdiction-specific gaming requirements. Analysed **150+ games in total** during the role.

EDUCATION

- 2024–2026 **MSc Artificial Intelligence**
 UNIVERSITY OF ESSEX · United Kingdom 
 Dissertation on internal representations in sequence models (see Research), supervised by Athanasios Chasapis.
- 2017–2018 **BSc Data Science & Knowledge Engineering (one year)**
 MAASTRICHT UNIVERSITY · Netherlands 
 Learned to program from zero. Selected grades: Data Structures & Algorithms **10/10**, Discrete Mathematics **9/10**, Computer Science I **9/10**. Teaching assistant in Mathematical Analysis. Left to join GLI Europe.
- 2012–2016 **MSc Theoretical Physics**
 UTRECHT UNIVERSITY · Netherlands 
 General relativity, string theory, group theory, statistical physics. Thesis combining statistical physics with game theory.
- 2006–2011 **BSc Physics**
 ARISTOTLE UNIVERSITY OF THESSALONIKI · Greece 
 Mathematics, probability, statistics, modelling, numerical reasoning.

SELECTED PROJECTS

- MSc **Emotion Recognition & Image Description App**
 ARTIFICIAL INTELLIGENCE · GRADE: **83/100**
 Flask app: MediaPipe face detection (via DeepFace), 7-class emotion recognition with a CPU-friendly mini-Xception (FER2013), age and gender estimation, image descriptions with Mistral-7B via the Hugging Face Hub; drives an emotion-based music player. Traded a little accuracy for **stable real-time inference on CPU**. Compared zero-, one- and few-shot prompting with BERTScore and perplexity (few-shot best; the metrics did not always agree).
 Live demo (Hugging Face Spaces)  Code
- MSc **Customer Churn with Neural Networks**
 NEURAL NETS & DEEP LEARNING · GRADE: **95/100**
 Telco dataset (7,043 customers, 26% churn). MLP in TensorFlow/Keras tuned with Optuna (TPE sampler, median pruning), benchmarked against grid-searched logistic regression and CatBoost. Class weights beat SMOTE (**F1 0.619 vs 0.584**); CatBoost best overall (**F1 0.625, ROC-AUC 0.836**); SHAP: contract type, monthly charges and tenure drive churn.
- MSc **Reddit Topic & Sentiment Analysis**
 ML & PYTHON · GROUP LEAD, TEAM OF 4 · GRADE: **88/100**
 End-to-end NLP pipeline: posts collected via the Reddit API across four themes, double annotation with majority vote, TF-IDF features and clustering, six classifiers for topic and sentiment, bigrams and sentiment-over-time analysis. Random Forest topic accuracy **0.748** (chance ≈ 0.25). Split tasks, coordinated annotation agreement, wrote the final report.